

Keyu Wang

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Education

Harvard University

Master of Science in Data Science

September 2025 – Present

Cambridge, MA, USA

McGill University, cGPA: 3.96/4.0

B.Sc. in Computer Science, minor in Geographic Information Science

September 2020 – December 2024

Montreal, QC, Canada

Courses: Applied Machine Learning, Natural Language Processing, Computer Vision, Database Systems, Algorithm Design, Discrete Mathematics, Statistics, Advanced Geographic Information Science

Awards & Scholarships

Undergraduate Student Research Award (USRA) 2024

\$8,700 | Issued by Natural Sciences and Engineering Research Council of Canada (NSERC)

Undergraduate Student Research Award (USRA) 2023

\$9,000 | Issued by NSERC

Research Assistantship Stipend

\$2,550 | Issued by Social Sciences and Humanities Research Council (SSHRC)

Perseverance Award for Women in Technology 2022

\$3,000 | Issued by Quebec Government

McGill University Dean's Honour List 2022

McGill University Dean's Honour List 2021

Publications

1. **K. Wang***, J. Li*, S. Yang, Z. Zhang, D. Wang. *When Truth Is Overridden: Uncovering the Internal Origins of Sycophancy in Large Language Models*. [Paper]
2. S. Yang, S. Zhu, Z. Wu, **K. Wang**, J. Yao, J. Wu, L. Hu, M. Li, D.F. Wong, D. Wang. *Fraud-r1: A Multi-round Benchmark for Assessing the Robustness of LLMs Against Augmented Fraud and Phishing Inducements*. In **Findings of the Association for Computational Linguistics (ACL), 2025**. [Paper]
3. **K. Wang**, A.N. Iranzad, S. Schaffter, M. Risdal, D. Precup, J. Lebensold. *Mitigating Downstream Model Risks via Model Provenance*. In **NeurIPS 2024 Workshop on Socially Responsible Language Modelling Research, 2024**. [Paper]

Work Experience

Machine Learning Engineer

Moonarch

Feb 2025 – Present

Remote

- Developed document extraction pipeline for startup company profiles, getting structured insights from unstructured documents, enabling accurate parsing of mixed-format business documents containing images and embedded text.

Research Intern

Okinawa Institute of Science and Technology (OIST)

May 2025 – August 2025

Okinawa, Japan

- Joined the Machine Learning and Data Science Unit under **Prof. Makoto Yamada**, focusing on fractal video pretraining on brain-inspired vision foundation model.

Visiting Student Researcher

King Abdullah University of Science and Technology (KAUST)

Jan 2025 – May 2025

Thuwal, Saudi Arabia

- Conducted research in the Provable Responsible AI and Data Analytics (PRADA) Lab under **Prof. Di Wang**.
- Explored LLM sycophancy behaviours, designing a prefix framework to test models like Llama 3.2 and Qwen 2.5 and conducted mechanistic analysis to explain the reason why.

Research Intern on AI Safety

Mila, AI Research Institute

May 2024 – December 2024

Montreal, QC, Canada

- Under the supervision of **Prof. Doina Precup** with collaborators from Google & Meta, illustrated model provenance risk in healthcare, identified key properties for early warning systems & proposed an open-source, community-led repository for tracking provenance to enhance transparency and establish a new standard for responsible model management.
- As the 1st author, [paper](#) is accepted to NeurIPS 2024 workshop: Socially Responsible Language Modelling Research.

Data Scientist Intern

May 2024 – August 2024

Bell Canada

Montreal, QC, Canada

- Conducted data analysis on **Google Cloud Platform**, enhancing service assurance outcomes by streamlining query performance and utilizing advanced RegEx to extract and refine data efficiently.
- Reduced the initial dataset to 6% of its original size, focusing on high-value data, and delivered actionable insights using effective data visualization, while collaborating within the team using Agile methodologies.

AI & Data Engineering Intern

May 2023 – December 2023

Bell Canada

Montreal, QC, Canada

- Built ML algorithms including clustering for Business Intelligence problems using **Teradata SQL**, Python ML frameworks such as **Scikit-learn**.
- Data analysis & engineering support for building training dataset of a Presence-only model about Competitors' Network Speed Analysis, following ETL pipelines.
- Utilized Agile methodologies (Scrum) to ensure effective collaboration, using **JIRA Board & Confluence**.

Undergraduate Research Assistant

February 2023 – January 2024

Department of Geography, McGill

Montreal, QC, Canada

- Built project website from scratch for Historical Geography research project under the supervision of **Prof. Raja Sengupta** and **Prof. Griet Vankeerberghen**, using JavaScript, HTML, Leaflet.
- Automated map digitization for boundary changes throughout 200 years in imperial China, using Python & ArcGIS.

Head Instructor Coding & Game Design

June 2022 – August 2022

Appleby College Camps

Oakville, ON, Canada

- Planned course materials for coding and video game design camps, and taught basic Python and GameMaker to over 140 children from 7 to 14 years old.
- Organized weekly showcases & maintained close communication with campers' families for study updates.

Projects

McGill School of Computer Science Scheduling Website | *Web Development*

December 2024

- As one of the 2 backend developers who made a user-friendly platform that simplifies appointment scheduling.

You Are At Where You Tweet: GPT Prompting to Geo-locate Twitter Users | *LLM, AI safety*

April 2024

- Employed a dataset featuring user-defined locations and tweet texts, the study involved grouping tweets by user, cleaning, and refining input prompts to optimize accuracy.
- Final top-3 accuracy of 47% for worldwide city location inference & 82% for Australian city predictions.

Frame Prediction for Aerial Objects | *Computer Vision, OpenCV, MATLAB*

December 2022

- Developed object tracking & frame prediction methods for aerial objects in MATLAB using classic (non-AI) computer vision algorithms.

California Gold Rush Historical Storymap | *JavaScript, HTML, MapBox, GeoJSON*

October 2022

- Visualized the history and impacts of immigration to California during 1848-1869 via a storymap.

Presentations

Socially Responsible Language Modelling Research (SoLaR) workshop at NeurIPS 2024

December 2024

- Given poster presentation about the "Mitigating Downstream Model Risks via Model Provenance" paper at NeurIPS workshop in Vancouver, Canada.

Undergraduate Computer Science Research Symposium (UCORE) | *LLM, AI safety*

September 2024

- Given poster presentation about the "Mitigating Downstream Model Risks via Model Provenance" project at a school-level research poster session with 60+ attendees.

GIS (Geographic Information Science) Day 2022 | *Historical Geography*

November 2022

- As an outstanding student project, invited to give an oral presentation at GIS Day hosted by McGill for presenting a story map visualizing the history and impacts of immigration to California during 1848-1869.

Skills

Programming Languages: Python, Java, C++, C#, HTML/CSS, JavaScript, MATLAB, Bash, OCaml

IDE/Tools: VS Code, Eclipse, IntelliJ IDEA, PyCharm, Jupyter Notebook

Technologies/Frameworks: Linux, Git, GitHub, GitLab, JUnit, JavaFX, Pandas, Sci-kit Learn, TensorFlow, PyTorch

Web Development : XAMPP, React, Node.js

Databases: PostGreSQL, DB2, Teradata SQL, Microsoft SQL, SAS

Video Production : Adobe Premiere Pro, Adobe Lightroom

Spoken Languages: English, Mandarin Chinese(both native fluency), Japanese (JLPT N2)

Leadership / Extracurricular

East2West

April 2023 – Present

Dancer, Editor

- Danced & edited long videos for a K-pop cover performance crew with a YouTube channel of 1.5 million subscribers, achieving on average 20K views per video.
- Participated in the “2024 K-pop Cover Dance Festival” and won 3rd place in the Canada division.

K-Rave McGill

September 2021 – December 2024

Dance Leader

- Led K-POP dance projects by organizing practice logistics, teaching dances, and giving members weekly individual feedback for improvement to ensure performance quality.

McGill University Chinese Students & Scholars Association

September 2020 – April 2024

Vice President Media - Senior Advisor

- Directed, filmed & edited creative promotion video content for club sponsors (such as Liuyishou Hotpot, Presotea Montreal, etc.) 3 times yearly as a part of the signed discounting contract, to benefit all McGill Chinese students from receiving 10 to 25% off discounts.
- Co-directed Montreal Lunar New Year Gala 2022 with 100+ performers and 15 staff, actively involved in venue booking, auditions, rehearsals, & stage setup.
- Developed webpages on WeChat Official Accounts for 70+ sponsor stores promotion displays; got 1000+ student users.
- Organized member photoshoot for 70 internal members, involving studio visiting & bookings, creative ideas, pose directing & photography.